9.1 Capturing and understanding the current state

Whenever someone is planning a major renovation of their home interior, they usually hire an interior designer. But before the designer can start work, they have to capture a current floor plan. To make meaningful decisions about what can be changed, they need to understand the current state. As an example, you might not be able to remove every wall, because you cannot easily change load-bearing walls. Only partition walls can be easily moved.

This planning process is similar to the work done by software architects. If you do not understand the systems that are already in place and how they are connected, you cannot plan any changes. This is why we will start this chapter by capturing the current state of the Messflix architecture. Furthermore, it will not be enough to understand the end state. We also need to understand the reasoning behind decisions that were made so we can make conscious choices about changes. We will also have to understand architectural drivers for our data products.

This section will show you the kind of architectural information you need to gather before jumping into the design of data products. If your systems are already well documented, good for you! This section will help you navigate through this documentation and look for pieces of information that matter. If you are not so lucky, and the complete documentation is in the heads of developers and architects, this section will help you capture that documentation. But before we do that, you need to understand what architecture is.

9.1.1 What is software architecture?

Let’s start by clarifying what we mean by *software architecture*. For us, architecture is the structure of the software and the process of designing it.

By *architecture* as a *structure*, we mean the building blocks of the system; the relations and interfaces between them; the patterns applied; the technology stack for the building blocks, platform, or infrastructure used by these components; and finally cross-cutting concerns like security and logging. By *architecture*, we mean elements of the structure that take time to change.